F L E E T & S H O R E P E R S O N N E L T R A I N I N G

Ship Handling Training

The forces that govern how a ship turns and stops, manoeuvring in open and confined waters, and safe berthing, anchoring and emergency turns.

IMO MSC.137(76)

STCW II/1 & II/2

Company SMS

COURSE MAP

What We Will Cover

01 **Manoeuvring Fundamentals**
Pivot point, squat and the forces that govern how a ship turns and stops

02 **Manoeuvring in Open Water**
Turning circles, stopping distances and the crash-stop manoeuvre

03 **Shallow & Restricted Water Effects**
Squat, bank effect and interaction in confined channels

04 **Tugs & Pilotage**
Working safely with tugs and embarked pilots

05 **Berthing & Unberthing**
Planning and executing alongside manoeuvres

06 **Anchoring & Emergency Manoeuvres**
Safe anchoring and the Williamson / Anderson / Scharnow turns

07 **The Bridge Team's Role**
Roles, communication and monitoring throughout the manoeuvre

SECTION 1

Manoeuvring Fundamentals

01

The physical forces — pivot point, squat and bank effect — that determine how every ship answers her helm.

HYDRODYNAMICS

How a Ship Actually Turns

Turning is governed by forces acting fore and aft of a moving pivot point, not by the rudder alone.

- The pivot point is the point about which the ship appears to rotate; for a vessel moving ahead it lies roughly a third of the length from the bow.
- Water flowing past the hull and rudder creates unequal pressure forces fore and aft of the pivot point, producing the turning moment.
- As speed increases, rudder force and turning response both increase, but so does the distance needed to check the swing once it has begun.
- The pivot point shifts aft toward amidships as the ship loses headway, and moves further aft still once the ship is moving astern.
- Trim, draught and loading condition all alter where the pivot point sits and how promptly the ship answers helm and engine orders.

Know Your Ship

Every vessel has her own manoeuvring characteristics — always consult the ship's pilot card and manoeuvring booklet before a critical evolution, not general rules of thumb.

CORE CONCEPTS

Four Effects Every Watchkeeper Must Understand

These effects combine, and intensify sharply as water gets shallower or channels narrower.

Pivot Point

The point of apparent rotation, shifting with speed and direction of travel — not the ship's centre of gravity.

Squat

The vessel sinks and trims as speed increases in shallow or confined water, reducing under-keel clearance without warning.

Bank Effect

A ship close to a channel bank is drawn toward it at the bow and pushed away at the stern, requiring continuous helm correction.

Interaction

Passing or overtaking close to another vessel or a fixed structure sets up suction and repulsion forces that can pull ships together.

WORKED NUMBERS

Why Squat Catches Crews Out

Squat is easy to underestimate, and its consequences appear suddenly, with little warning.

2-3x

Squat at high block-coefficient, shallow-water speed can be two to three times deeper than a simple rule-of-thumb estimate

UKC

Under-keel clearance must account for squat, trim, swell and chart datum error, not static draught alone

V^2

Squat grows roughly with the square of speed — a modest speed reduction gives a disproportionately large reduction in squat

SECTION 2

02

Manoeuvring in Open Water

Turning circles, stopping distances and the crash-stop manoeuvre — baseline data every bridge team should know before an emergency.

MANOEUVRING DATA

Reading the Turning Circle

IMO standards require every ship to carry manoeuvring data derived from full-scale sea trials.

- Advance is the distance travelled in the original heading direction from the moment the helm is applied to the point the heading has changed 90 degrees.
- Transfer is the distance travelled at right angles to the original course over the same 90-degree turn, showing how far the ship has moved sideways.
- Tactical diameter is the distance between the original course line and the ship's course when she has turned through 180 degrees — the practical measure of turning room.
- IMO Resolution MSC.137(76) sets minimum manoeuvring standards; results for the ship's own hull and loading conditions appear on the pilot card and wheelhouse poster.
- Turning characteristics vary significantly between ballast and laden condition, and between calm water and a seaway — trial data is a guide, not a guarantee.

Pilot Card

The pilot card summarising turning circle, stopping distances and manoeuvring particulars must be handed to the pilot at every embarkation — check it is current.

MANOEUVRING DATA

Stopping Distances by Manoeuvre

Approximate distances vary with hull form, loading and engine type — always use the ship's own trial data.

Manoeuvre	Typical Track Distance	Key Consideration
Emergency (crash) stop from full sea speed	10-15 ship lengths	Full ahead to full astern; a large sheer off track is normal
Stop engine, allow the ship to drift down	Several times the crash-stop distance	Less sheer, but far longer distance and less control
Minimum stopping distance, laden large tanker	Up to 3-4 nautical miles	Reinforces the need for very early speed reduction in traffic

EMERGENCY MANOEUVRE

Executing a Crash Stop Safely

A full emergency stop is a last resort — it is unpredictable and puts real load on the ship and machinery.

- Order full astern only when collision or grounding cannot otherwise be avoided; the manoeuvre produces significant, often unpredictable, sheer.
- Transverse thrust from the propeller in astern power will swing the bow, typically to port for a right-handed single propeller — anticipated, not fought.
- Engine control must confirm the order is understood and executed; a delayed or partial response changes the stopping distance dramatically.
- Rudder should normally stay amidships during a crash stop unless a specific avoiding action requires otherwise, to avoid compounding the sheer.
- Every crash stop, planned or actual, should be logged and reported so the manoeuvring data can be reviewed against the ship's own trial figures.

SECTION 3

Shallow & Restricted Water Effects

03

Confined channels, canals and shallow approaches change the rules — squat, bank effect and interaction all intensify.

CONFINED WATER

Manoeuvring in Channels and Canals

Reduced under-keel and lateral clearance change the ship's response and raise the consequences of any error.

- In shallow water the ship's effective added mass increases, response to helm slows, and turning circles grow noticeably larger than in deep water.
- Bank effect grows sharply as the ship moves off the channel centreline; correcting too late requires more rudder and more time than correcting early.
- Speed should be reduced in good time before entering a restricted channel, both to control squat and to leave a margin for unexpected traffic.
- Two-way traffic in a narrow channel compounds bank effect with interaction between the passing ships, requiring reduced speed and early communication.
- Local pilotage information, tide tables and the latest notices to mariners must be reviewed before any restricted-water passage is attempted.

Squat Reserve

Always plan under-keel clearance with a squat and safety margin built in, not just static draught against charted depth — conditions can change during the passage.

BEST PRACTICE

In Narrow Channels and Canals

A few disciplined habits keep confined-water transits predictable.

Do

- Reduce speed early and steadily, well before entering the restricted section
- Keep to the centre of the channel where traffic and depth allow
- Brief the bridge team on expected bank and interaction effects beforehand
- Monitor under-keel clearance continuously against tide and squat allowance

Don't

- Don't attempt large course changes at full transit speed in a narrow channel
- Don't assume a wide margin exists just because deep water shows nearby on the chart
- Don't pass close to another vessel without briefing the team on interaction risk
- Don't delay reducing speed until squat or bank effect is already being felt

CLOSE-QUARTERS EFFECTS

What Happens When Ships Pass Close

Interaction forces are easy to underestimate until two ships are already being pulled together.

Suction

As vessels draw abeam, bow-out / stern-in suction can pull the ships together, strongest near the closest point of approach

Repulsion

As bows meet, a cushion of water pushes the bows apart before suction takes over amidships and astern

$Speed^2$

Interaction forces increase sharply with speed and reduce with lateral separation — slow down and open the distance

SECTION 4

Tugs & Pilotage

04

Working safely and effectively with embarked pilots and assisting tugs.

The Master/Pilot Relationship and Exchange

Embarking a pilot does not transfer the master's overriding authority and responsibility for the safety of the ship.

- The pilot advises on local conditions, traffic and passage; the master and bridge team retain full responsibility for the safe navigation of the vessel.
- A Master/Pilot Information Exchange (MPX) must be completed before the pilot takes the conn, covering the passage plan, manoeuvring characteristics and any defects.
- The bridge team must monitor the pilot's actions and orders exactly as it would the master's own, and challenge any order that appears unsafe.
- Pilot card and wheelhouse poster data should be handed over and briefly discussed, so the pilot understands this particular ship's handling characteristics.
- If the master has doubts about the pilot's actions, the master has the right and duty to intervene, request clarification, or take over the conn.

Never Silent

A questioning attitude toward the pilot is not a lack of trust — it is part of the safety net every bridge team is expected to provide, always.

Tug Types and Making Fast Safely

The right tug, correctly made fast, is one of the most powerful safety margins a manoeuvre can have.

Conventional Tug

Pushes or tows on a line from the bow, stern or beam; power and manoeuvrability are lower than azimuth types.

Azimuth Stern Drive (ASD) Tug

Rotating thrusters give power and control in any direction, widely used for close escort and berthing work.

Making Fast

Agree the method, line and communication channel with the tug master before starting; never exceed the tug's safe working load.

Emergency Tow-Line Release

Every crew handling tug lines must know the quick-release arrangement and keep clear of the line's bight and snap-back zone.

PLANNING

Choosing Tug Numbers and Positions

The number and position of tugs depends on ship size, windage, berth layout and available power.

Situation	Typical Tug Use	Key Risk
Large ship, tight berth, high windage	Two or more tugs, bow and stern	Insufficient power to hold position against wind or current
Moderate size, sheltered berth	Single tug, made fast forward or aft	Delay in tug response if conditions change quickly
Escort through a restricted channel	Tug tethered astern, standing by	Line-handling risk if emergency steering assistance is called for

SECTION 5

Berthing & Unberthing

05

Turning a passage plan into a safe approach, a controlled alongside manoeuvre, and a clean departure.

PLANNING

Planning the Approach to the Berth

A well-planned approach gives the ship time and sea room to correct for the unexpected.

- The berthing plan should be briefed to the whole bridge and mooring team, including approach angle, planned speed, wind/current allowance and abort points.
- Approach speed should be reduced in good time so the ship arrives at the berth with only steerage way, using minimum engine movements for the final approach.
- Wind and current on the beam must be assessed and allowed for, often by approaching at a small angle into the set, correcting progressively as the berth nears.
- Mooring lines should be run in a planned sequence — typically headline and stern line first to check the ship's way, then springs to control surge.
- An abort plan — a point beyond which the master will wave off and re-approach rather than continue — should be agreed before starting the approach.

Berth-to-Berth Brief

Passage and berthing plans should be briefed to the full team before arrival, not improvised on approach — surprises at the berth cost time and margin.

EXECUTION

During the Final Approach

Discipline in the last few ship-lengths determines whether the berthing is routine or eventful.

Do

- Keep engine movements small and confirm each order is executed correctly
- Use tugs and mooring lines progressively rather than relying on engine alone
- Maintain continuous communication between bridge, mooring stations and tugs
- Monitor rate of closure to the berth constantly, not only at set intervals

Don't

- Don't approach faster than the ship can be stopped safely in the remaining distance
- Don't commit to the berth if wind or current has changed materially since planning
- Don't let mooring parties work near a line that is under load or in its snap-back zone
- Don't dismiss the tugs or release lines until the master confirms the ship is secure

DEPARTURE

Leaving the Berth in Control

Unberthing deserves the same planning and discipline as the approach.

Order

Lines are normally let go in reverse sequence to the berthing plan, springs first, to keep the ship under control throughout

Set & Drift

Wind and current off the berth can assist an unberthing, or must be countered with tugs and engine if pushing the ship onto it

Check

Confirm clearance from adjacent vessels, moorings and underwater obstructions before ordering any engine movement

SECTION 6

Anchoring & Emergency Manoeuvres

06

Safe anchoring practice, and the standard turns every bridge team must be ready to execute without hesitation.

Planning and Letting Go the Anchor

Anchoring looks routine until wind, current or poor holding ground turns it into an emergency.

- Select an anchorage with adequate depth, swinging room and holding ground, checking the chart for foul ground, cables and other anchored traffic.
- Approach into the wind or current, whichever is dominant, reducing to minimum steerage way before letting go to avoid excessive cable run-out.
- Normal scope is typically four to six times the depth of water in moderate conditions, increased in strong wind, exposed anchorages or poor holding ground.
- Once brought up, confirm the anchor is holding by taking bearings or GPS position fixes at regular intervals, watching for a dragging anchor.
- An anchor watch must be maintained continuously, ready to raise the alarm, start the main engine and take avoiding action if the anchor begins to drag.

Dragging Anchor

A steady change in bearing of fixed shore objects, or a GPS position moving outside the swinging circle, are the first signs of a dragging anchor — act immediately.

STANDARD MANOEUVRES

Turns Every Officer Must Know Cold

Man-overboard recovery turns are among the very few manoeuvres a bridge team must be able to execute from memory.

Williamson Turn

Brings the ship back onto her reciprocal track through the point where a person was lost — the standard turn when the casualty was not seen to fall.

Anderson Turn

A fast single-turn recovery bringing the ship back to the casualty by the shortest route, suited to smaller, highly manoeuvrable vessels in good conditions.

Scharnow Turn

Used when some time has elapsed since the person went overboard, retracing the ship's own wake efficiently over a longer interval.

Immediate Action

Full rudder toward the side the person fell, engine as required, lifebuoy and marker released, lookout kept on the casualty throughout the turn.

DECISION AID

Matching the Manoeuvre to the Situation

The right turn depends primarily on how much time has passed and whether the casualty is in sight.

Situation	Recommended Turn	Why
Casualty seen to fall, still in sight	Immediate action / single turn	Fastest possible return while contact is maintained
Casualty not seen to fall, position uncertain	Williamson turn	Returns the ship precisely onto the original track to search
Some time elapsed since the person went missing	Scharnow turn	Efficiently retraces the track over a longer time interval

SECTION 7

The Bridge Team's Role

07

Manoeuvring is a team evolution — clear roles, closed-loop orders and continuous monitoring keep it safe.

BRIDGE ORGANISATION

Who Does What During a Manoeuvre

Every manoeuvring evolution should have clearly assigned roles, briefed before it begins.

- The conning officer, often the master or pilot, gives helm and engine orders and holds overall situational responsibility for the manoeuvre.
- The helmsman repeats every order back before executing it, and reports when the ordered heading or rudder angle has been reached.
- A dedicated navigator monitors position, speed and under-keel clearance independently, cross-checking the conning officer's mental picture.
- A lookout is maintained throughout, particularly in restricted visibility or heavy traffic, kept separate from anyone occupied with orders or plotting.
- Engine control confirms each telegraph or engine order and reports promptly if the engine cannot achieve the ordered movement.

Closed-Loop Communication During Manoeuvres

The manoeuvre is only as safe as the communication that runs through it.

Do

- Use standard, unambiguous helm and engine orders every time
- Require every order to be read back before it is executed
- Speak up immediately if the ship's response does not match the order given
- Keep a running commentary of position and clearance during critical phases

Don't

- Don't assume an order was understood just because it was acknowledged quietly
- Don't let radio, phone or unrelated conversation distract the conning officer
- Don't hesitate to challenge the master or pilot if something looks wrong
- Don't allow one person's mental picture to go unchecked during the manoeuvre

SITUATIONAL AWARENESS

What the Team Should Be Tracking

Three threads of awareness should run continuously through any manoeuvre.

Position

Continuous position fixing against the plan, with a clear trigger for when the ship is judged to be off track

UKC

Under-keel clearance checked against tide, squat and the minimum margin set in the passage plan

Traffic

Other vessels, tugs and shore structures tracked visually and on radar/AIS throughout the evolution

COURSE COMPLETE

Confident, well-briefed manoeuvring — not chance — is what keeps a ship, her crew and the port around her safe.

Company DPA — 24/7 emergency line

Fleet Marine Operations Department — mng@gloryship.com.tr

Fleet Safety Department — mng@gloryship.com.tr